2019A F=ma Exam Review

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Problem 3

Use momentum conservation (essentially a perfectly inelastic collision). Use energy conservation (no external forces) and consider convenient points: initial kinetic = final kinetic of both (moving at the same speed) $+ mgh$. You can write K as $p^2/2m$ to avoid doing $mv = (M+m)v'$, because momentum is conserved (it stays mv for the system). p is the total momentum and m is the total mass, so

$$K = \frac{(mv)^2}{2(M+m)}.$$

Problem 4

Problem-solving strategy: instead of thinking there's not enough info, what if you can solve for a ratio instead? $T \propto \sqrt{m/k}$. We have neither m nor k, but we can easily find an expression for m/k using basic energy conservation. In SHM, velocity $= 0$ implies an extreme. The equilibrium point is where $mg = kx$. When you release it, at one point it passes the equilibrium point. If it travels $2x$, then the upward force is $2mg$ and the net force is mg upward, which proves the top of the motion is the release point and the equilibrium point is halfway down. So $mg = kx$, $m/k = x/g$, and (subbing that in for m/k)

$$T = 2\pi\sqrt{\frac{x}{g}}.$$

Problem 6

Work in the frame of the plane. Rotating your coordinates shows that in the y direction (perpendicular to the incline) the ball simply collides elastically (reversing v_y when hitting the stationary plane) and bounces back to the same height (perpendicular distance from the incline), conserving the y-motion energy, while being accelerated in the x direction by gravity.

Problem 8

Pay attention to the fact that the problem states the earth has radius R and the object is at a distance R from the surface. So the r in the circular orbit and escape v's would be $2R$.

$$v_c = \sqrt{\frac{GM}{r}}, \qquad v_{\text{esc}} = \sqrt{\frac{2GM}{r}}.$$

$\sqrt{2GM/3R} = \sqrt{4GM/3r}$ is between circular and escape, so it'll be in an elliptical orbit starting closest to earth. (If $\sqrt{GM/r} > v > 0$, then the object has an elliptical orbit starting furthest from earth.)

Problem 9

Draw the vectors, do the math, and use the half-angle identity

$$\sin\frac{\theta}{2} = \sqrt{\frac{1-\cos\theta}{2}}.$$

Problem 10

The question said "if the remaining shape has mass M," which implies the full disk has more than M. To avoid arithmetic errors, set the original to m and leave the final as M, then convert from M to m. Similarly, for other problems, feel free to create your own variables and convert at the end.

Problem 12

This is a challenging math-related problem. Identify the goal; this is very important. Know clearly what you're trying to find: the time between bounces τ as a function of the total time t. The total time is found via the sum of a finite geometric series:

$$\frac{a(1-r^n)}{1-r}.$$

You'll be able to solve for the time for the nth bounce, $\tau(n) = \tau\alpha^n$, in terms of t. Combining the total time and the time for the nth bounce eliminates n, and you see the linear relationship between $\tau(n)$ and t.

Problem 14

Coriolis acceleration $= -2\,\boldsymbol{\omega} \times \mathbf{v}$. Use $d = \frac{1}{2}at^2$ for the deflection and $x = vt$ for the distance travelled, both in the rotating frame.

Problem 15

Period of rotation = time in the air. Since the rod is initially upright (vertical), the initial horizontal velocity is the standard tangential $v = r\omega$, where r is the distance to the com.

Problem 18

Key idea: work done by the launcher $= \Delta K$ for the wedge. OR: initial potential stored in the launcher = final K of the wedge. Whether you approach it with the work-energy theorem or energy conservation depends on what you include in your system.

Problem 19

The easiest way is to compare the new to the old restoring force (mg) to get $g_{\text{eff}} = g/2$, and use $\omega = \sqrt{g_{\text{eff}}/L}$.

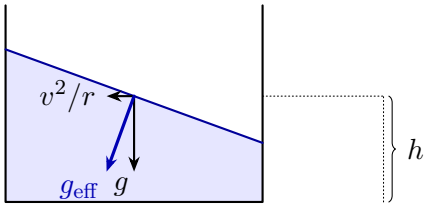
Problem 20

Balance the torques. For stability, the restoring torque must be greater than the destabilizing torque. $dx = L\,d\theta$.

Problem 24

$F = -dU/dx$. Once you have the restoring force, you can get ω and T. T_{min} is the shorter period of x or y, which occurs if the particle only moves in that dimension (and isn't moving in the other). T_{max} occurs when there's oscillation in both dimensions, and the period only resets once they synchronize again, so it must be $\text{LCM}(T_x, T_y)$. (The foundation is that the function $U(x, y)$ has no cross terms like xy, so we can treat the x and y directions independently.)

Problem 25

Bernoulli: $\rho g h = \frac{1}{2}\rho v^2$. In the rotating frame when turning, there is a centrifugal acceleration v^2/r, so g_{eff} is at an angle to the vertical. Therefore the water surface will tilt, and the "effective" height also isn't just h. Geometry: $h_{\text{eff}} = h\cos\theta = h\,\dfrac{g}{g_{\text{eff}}}$, so

$$v = \sqrt{2 g_{\text{eff}} h_{\text{eff}}} = \sqrt{2gh}.$$